Molecules and solids

[END EXAM - 2023/2024]

(a)(i) Primary: Ionic, metallic, covalent
secondary: Hydrogen bond, Dipole-dipole (Keesom)
Dipole-induced dipole (Debye), London dispersion.

(ii). Primary bonds are strong. They form by transferring or sharing valence electrons (ionic, covalent) or by delocalising them (metallic).

Secondary bonds are much weaker. They form by attraction between permanant or temporary dipoles. Involve no transfer or sharing e⁻s.

• primary bonds: Holds atoms together within a molecule or solid, secondary bonds: Acts between molecules or chain.

(b)(i). • Melting point / boiling point (bond strength)
• Elastic (young's) modulus / stiffness
• Thermal expansion coefficient : Also { p. Bond length ; strength

(ii) Melting/boiling point: The depth of the well is the bond energy. A deeper well means more energy needed to seperate the atoms, so high melting and boiling points.

Thermal expansion: The curve is asymmetric: steep repulsive side and a shallow attractive side. As T↑ atoms vibrate with more energy and the mean position shifts to larger r. ↑ narrow, deep more symmetric well gives less expansion.

stiffness: The modulus is proportional to the curvature at the minm, $E \propto (d^2u/dr^2)$ at r_0, steeper, narrower means larger curvature, E↑ stiffer.

(ii) X_2 has its minm at smallest internuclear distance
Shortest bond. ⇒ H_2
Z_2 has the deepest well so the strongest bond. ⇒ N_2
Y_2 is intermediate compared to other O_2 ⇒ O_2

(c) $V(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$

(i) Repulsive potential : $V_{rep} = 4\varepsilon\left(\frac{\sigma}{r}\right)^{12}$

origin : Dominates at very short distances. Arises from Pauli exclusion principle as atoms forced to close together their electron clouds overlap ⇒ strong repulsive force.

Attractive potential : $V_{attr} = -4\varepsilon\left(\frac{\sigma}{r}\right)^{6}$

origin : Dominates at longer ranges. Represents van der waals forces. London forces caused by interacting instantan-eous dipolemoment

(ii) At Equilibrium:

$$\left.\frac{dV}{dr}\right|_{r=r_0} = 0$$

$$V(r) = 4\varepsilon(\sigma)^{12}(r^{-12}) - 4\varepsilon(\sigma)^{6}(r^{-6})$$

$$\frac{dV}{dr} = 4\varepsilon(-12\sigma^{12}r^{-13} + 6\sigma^{6}r^{-7})$$

$$0 = 4\varepsilon(-12\sigma^{12}r^{-13} + 6\sigma^{6}r^{-7})$$

$$2\sigma^{12}r^{-13} = \sigma^{6}r^{-7}$$

$$\sigma^6 = \frac{1}{2}r_0^6 \Rightarrow \sigma = \left(\frac{1}{2}\right)^{1/6} r_0$$

(iii) $r_0 = \sigma(2)^{1/6}$
$= 1.5808 \times 2^{1/6}$
$r_0 \approx 1.774\,\text{Å}$

(iv) $K = \left.\frac{d^2V}{dr^2}\right|_{r=r_0}$

$$\frac{d^2V}{dr^2} = 4\varepsilon(-12\cdot-13\,\sigma^{12}r^{-14} + 6(-7)\sigma^{6}r^{-8})$$

$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} = \frac{(6.626\times10^{-34})(3\times10^{8})}{1.955\times10^{-23}\ J}$

$\lambda \approx 1.017\times10^{-2}\ m$

(iii) Isotope ^{37}Cl has a greater atomic mass than ^{35}Cl
Higher μ, (same bond length - given) $\sim p\ I = \mu r^2$
$E \propto \frac{1}{I}$ ↓ ↑ $\lambda \propto \frac{1}{E}$ ↓ $\Rightarrow$ Higher wavelength. (longer)

(b)(i) $E_n = (n+\frac{1}{2})\frac{h}{2\pi}\sqrt{\frac{K}{\mu}}$

$\Delta E = E_{n+1} - E_n = [(n+1+\frac{1}{2}) - (n+\frac{1}{2})]\frac{h}{2\pi}\sqrt{\frac{K}{\mu}}$

$\Delta E = \frac{h}{2\pi}\sqrt{\frac{K}{\mu}}$

E_n = Energy of the nth vibrational state
n = vibrational quantum number
h = planck's constant
K = Effective spring constant
μ = reduced mass of the diatomic molecule.

(ii) $E = \frac{1}{2}KA^2$, $K = 323\ Nm^{-1}$

$E_1 = (1+\frac{1}{2})\frac{h}{2\pi}\sqrt{\frac{K}{\mu}} = \frac{3}{2}\times\frac{6.626\times10^{-34}}{2\pi}\sqrt{\frac{323}{2.9024\times10^{-26}\ kg}}$

$= 1.67\times10^{-20}\ J$

$E_1 = \frac{1}{2}KA^2 \Rightarrow \sqrt{\frac{2E_1}{K}} = A \Rightarrow \sqrt{\frac{2\times1.67\times10^{-20}}{323}} \approx 1.017\times10^{-11}\ m$

$A = 0.1017\ A^{\circ}$

$\frac{d^2V}{dr^2} = 4\varepsilon\left(156\sigma^{12}r^{-14} - 42\sigma^6 r^{-8}\right)$

$r = r_0$

$\frac{d^2V}{dr^2} = \frac{4\varepsilon}{r_0^2}\left(156\left(\frac{\sigma}{r_0}\right)^{12} - 42\left(\frac{\sigma}{r_0}\right)^6\right)$

from, $\frac{\sigma}{r_0} = \left(\frac{1}{2}\right)^{1/6}$ or $\left(\frac{\sigma}{r_0}\right)^6 = \frac{1}{2}$, $\left(\frac{\sigma}{r_0}\right)^{12} = \frac{1}{4}$

$\frac{d^2V}{dr^2} = \frac{4\varepsilon}{r_0^2}\left(156\left(\frac{1}{4}\right) - 42\left(\frac{1}{2}\right)\right)$

$k = \frac{18.48\varepsilon}{r_0^2} = \frac{72\varepsilon}{r_0^2}$

(b) (a) (i) $E_{rot} = \frac{h^2}{8\pi^2 I} J(J+1)$

$I = \mu r^2$

Ch$_2$ ^{35}Cl$_2$ $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m^2}{2m} = \frac{m}{2}$

$m = 34.968853 \times 1u$

$r = 1.98 Å = 1.98 \times 10^{-10}$ m

$\mu = \left(\frac{34.968853 \times 1.66 \times 10^{-27} kg}{2}\right) = 2.9024 \times 10^{-26} kg$

$I = \mu r^2 = (2.9024 \times 10^{-26}) \times (1.98 \times 10^{-10})^2$

$I = 1.1378 \times 10^{-45} kgm^2$

$E_{rot} = \frac{h^2}{8\pi^2 I} = \frac{(6.626 \times 10^{-34})^2}{8\pi^2 (1.1378 \times 10^{-45})^2} = 4.887 \times 10^{-24} J$

$E_1 = (4.887 \times 10^{-24}) 1(1+1) = 9.774 \times 10^{-24} J$

$E_2 = (4.887 \times 10^{-24}) 2(2+1) = 2.932 \times 10^{-23} J$

(ii) $\Delta E = E_2 - E_1$

$= 2.932 \times 10^{-23} - 0.9774 \times 10^{-23}$

$= 1.955 \times 10^{-23} J$

$K = 169\ Nm^{-1}$ $m \approx 126.905\ u$ of I_2

$m \approx 35\ u$ of Cl_2

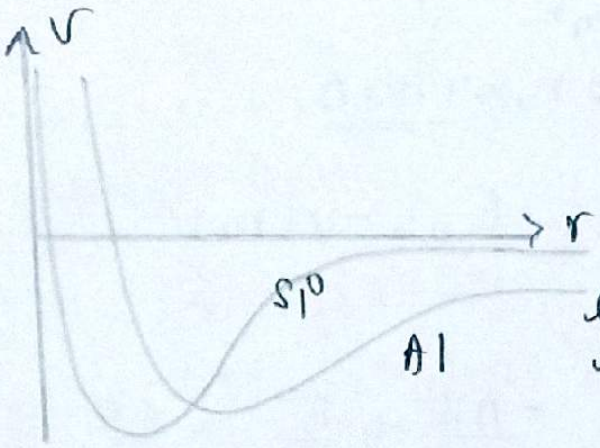
$E \propto \sqrt{\frac{K}{\mu}}$

Cl_2 has lower K and higher μ so $\sqrt{\frac{K}{\mu}}$ is higher for Cl_2 than I_2

So $^{35}Cl_2$ will absorb higher energy photon

[ENT EXAM 2022-2023]

(07)(a)



low melting point
low stiffness
higher thermal expansion

(b)(i) $V(r) = \frac{A}{r^{12}} - \frac{B}{r^6}$

Attractive potential: $-\frac{B}{r^6}$

origin: Dominates at longer distances. Represent vander waals forces. London forces caused by interacting instantaneous dipolemoment.

Repulsive potential: $\frac{A}{r^{12}}$

origin: Dominates at very short distances, Arises from pauli exclusion principle as Atoms are forced to close together their electron clouds overlap ∴ repulsive force. strong

(ii) $\left.\frac{dV}{dr}\right|_{r=r_0} = A(-12)r^{-13} + B.6\,r^{-7}$

$\frac{dV}{dr} = -12Ar_0^{-13} + 6Br_0^{-7} = 0$

$12A\,r_0^{-13} = 6Br_0^{-7}$

$2\frac{A}{B} = r_0^6 \rightsquigarrow r_0 = \left(\frac{2A}{B}\right)^{1/6}$

$r_0 = \left(\frac{2 \times 0.755 \times 10^{-119}\ eVm^{12}}{0.853 \times 10^{-59}\ eVm^6}\right)^{1/6}$

$r_0 \approx 1.10 \times 10^{-10}\ m \rightsquigarrow r_0 = 1.10\ \text{Å}$

(iii) $E_0 = -V(r_0)$

$E_0 = \frac{-A}{r_0^{12}} + \frac{B}{r_0^6}$

$= \frac{-A}{\left(\frac{2A}{B}\right)^2} + \frac{B}{\left(\frac{2A}{B}\right)} = \frac{-AB^2}{4A^2} + \frac{B^2}{2A} = \frac{B^2}{4A}$

$E_0 = \frac{(0.853 \times 10^{-59})^2}{(4 \times 0.755 \times 10^{-119})} \approx 2.41\ eV$

(iv) $\frac{d^2V}{dr^2} = -12(-13)Ar_0^{-14} + 6(-7)Br_0^{-8}$

$= 156Ar_0^{-14} - 42Br^{-8}$

$= \frac{156A}{r^{14}} - \frac{42B}{r^8}$ $\quad \because A = \frac{Br_0^6}{2} \Rightarrow r_0^6 = \frac{2A}{B}$

$\frac{d^2V}{dr^2} = \frac{156}{r_0^{14}}\left(\frac{Br_0^6}{2}\right) - \frac{42B}{r_0^8}$

$K = \frac{78B}{r_0^8} - \frac{42B}{r_0^8} = \frac{36B}{r_0^8}$ $\rightsquigarrow$ $= 2294\ N/m$

$K = \frac{36 \times 0.853 \times 10^{-59}}{(1.1001 \times 10^{-10})^8} \approx 14.32 \times 10^{21}\ eV/m^2$

$K = 14.32 \times 10^{21} \times 1.602 \times 10^{-19}$

(a) longest wavelength $J=0 \rightarrow J=1$

$$\Delta E = E_1 - E_0 = \frac{h^2}{8\pi^2 I}(1)(2) - 0 = \frac{h^2}{4\pi^2 I}$$

$$\mu = \frac{m_H \cdot m_{Cl35}}{(m_H + m_{Cl35})} = \frac{(1.007825u)(34.9688534u)}{(1.007825u) + (34.9688534u)}$$

$$= 0.979594u$$

$$\mu = 0.979594 \times 1.66 \times 10^{-27} kg$$

$$\approx 1.626 \times 10^{-27} kg$$

$$I = \mu r^2 \quad \therefore r = 0.12746 \times 10^{-9} m$$

$$= (1.626 \times 10^{-27})(0.12746 \times 10^{-9})^2$$

$$\approx 2.642 \times 10^{-47} kgm^2$$

$$\Delta E = \frac{(6.626 \times 10^{-34})^2}{4\pi^2 (2.642 \times 10^{-47})} \approx 4.210 \times 10^{-22} J$$

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{4.210 \times 10^{-22}} \approx 4.722 \times 10^{-4} m$$

(b) μ for ^{37}Cl $\quad m = 36.965903u$

$$\mu_{37} = \frac{(1.007825u)(36.965903u)}{(1.007825u + 36.965903u)} = 0.981084$$

$$= 0.98108 \times 1.66 \times 10^{-27} \approx 1.629 \times 10^{-27} kg$$

$$I = (1.629 \times 10^{-27})(0.12746 \times 10^{-9})^2$$

$$\approx 2.646 \times 10^{-47} kgm^2 \Rightarrow \Delta E \Rightarrow \lambda_{37} \approx 4.729 \times 10^{-4} m$$

(c) (i) Doublet: Because of these isotopes have dif atomic masses, $H-^{37}Cl$ and $H-^{35}Cl$ have slightly diff ed masses → diff I → diff E → diff λ

So these have diff λ but for same rotational transi so doublet occurs.

(ii) Δλ : $\Delta\lambda = |\lambda_{37} - \lambda_{35}|$

$= (4.7294 - 4.7223) \times 10^{-4}$

$= 7.200 \times 10^{-7} m$

(d) $\Delta E = \frac{h}{2\pi}\sqrt{\frac{E}{\mu}} = \frac{6.626 \times 10^{-34} Js}{2\pi}\sqrt{\frac{480\ Nm^{-1}}{1.6\ 28 \times 10^{-27} kg}}$

$\Delta E = (1.0546 \times 10^{-34} J\,s)(5.433 \times 10^{14} s^{-1})$

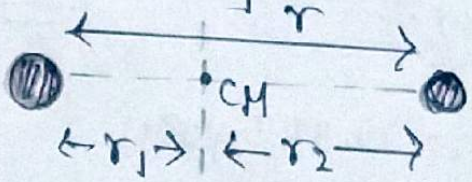
$\approx 5.730 \times 10^{-20} J$

$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34}) Js\ (3 \times 10^{8} m/s)}{(5.730 \times 10^{-20} J)}$

$\lambda \approx 3.469 \times 10^{-6} m$

END EXAM 2021/2022

07) (a)



$r = r_1 + r_2$

At CM: $m_1 r_1 = m_2 r_2$

$m_1 r_1 = m_2 (r - r_1)$

$(m_1 + m_2) r_1 = m_2 r$

$r_1 = \frac{m_2}{(m_1 + m_2)} r$ ∫ similarly $r_2 = \frac{m_1}{(m_1 + m_2)} r$

$= m_1 r_1^2 + m_2 r_2^2$

$= m_1 \left(\frac{m_2}{m_1+m_2} r\right)^2 + m_2 \left(\frac{m_1}{m_1+m_2} r\right)^2$

$I = \frac{m_1 m_2^2 r^2}{(m_1+m_2)^2} + \frac{m_2 m_1^2 r^2}{(m_1+m_2)^2} = \frac{m_1 m_2 r^2 (m_2+m_1)}{(m_1+m_2)^2}$

$I = \left(\frac{m_1 m_2}{m_1+m_2}\right) r^2$

$I = \mu r^2$ ■

(b) $E_{v,J} = (v+\frac{1}{2})hf + \frac{h^2}{8\pi^2 I} j(j+1)$

(i) $(v+1/2)hf$: Vibrational Energy ; v = vibrational quantum number

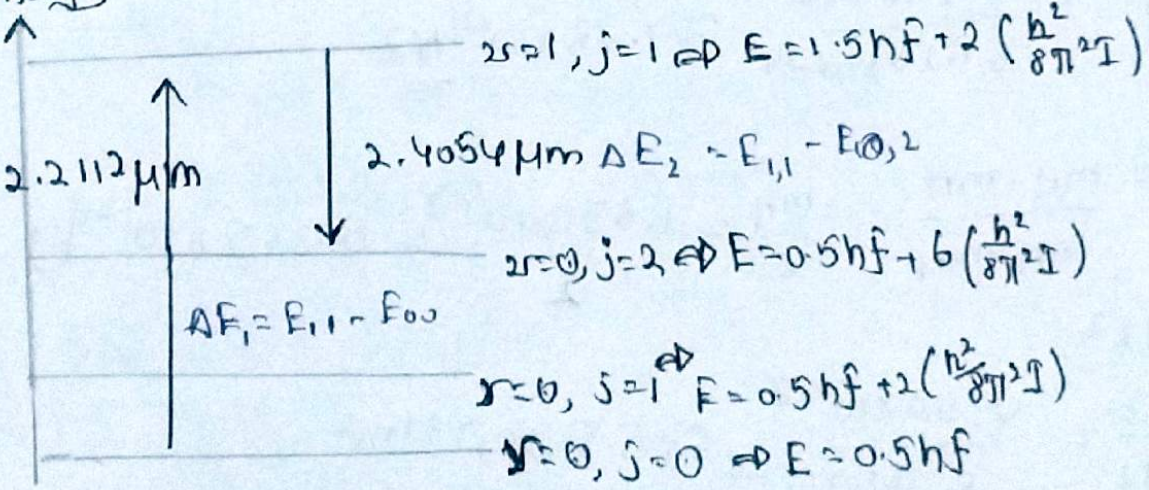
$\frac{h^2}{8\pi^2 I} j(j+1)$: Rotational Energy ; J = rotational quantum number

(ii) Initial state : $v=0, j=0 \Rightarrow E_{0,0} = \frac{1}{2}hf$

Intermediate state : $v=1, j=1 \Rightarrow E_{1,1} = 3/2\,hf + 2\left(\frac{h^2}{8\pi^2 I}\right)$

Final state : $v=0, j=2 \Rightarrow E_{0,2} = \frac{1}{2}hf + 6\left(\frac{h^2}{8\pi^2 I}\right)$

(i)



(2) $\Delta E_1 = E_{1,1} - E_{0,0} = hf + 2\left(\frac{h^2}{8\pi^2 I}\right) = \frac{hc}{\lambda_1}$ — (1)

$\Delta E_2 = E_{1,1} - E_{0,2} = hf - 4\left(\frac{h^2}{8\pi^2 I}\right) = \frac{hc}{\lambda_2}$ — (2)

(1) − (2)

$B = \frac{h^2}{8\pi^2 I}$

$(hf + 2B) - (hf - 4B) = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2}$

$6B = hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)$

$6B = (6.626\times10^{-34})(3.00\times10^{8})\left(\frac{1}{2.2112\times10^{-6}} - \frac{1}{2.4054\times10^{-6}}\right)$

$B = 1.210\times10^{-21}\,J$

$\frac{h^2}{8\pi^2 I} = B \Rightarrow I = \frac{h^2}{8\pi^2 B} = \frac{(6.626\times10^{-34})^2}{8\pi^2(1.210\times10^{-21})} = \frac{4.390\times10^{-67}}{95.53\times10^{-21}}$

$I \approx 4.60\times10^{-48}\,kgm^2$

(3) $hf + 2B = \frac{hc}{\lambda_1}$

$hf = \frac{1.9878\times10^{-25}}{2.2112\times10^{-6}} - 2(1210\times10^{-21})$

$f = \frac{8.748\times10^{-20}}{6.626\times10^{-34}} \approx 1.32\times10^{14}\,Hz$

(4) $\mu = \frac{m_H \cdot m_H}{m_H + m_H} = \frac{m_H}{2} = \frac{1.673\times10^{-27}}{2} = 8.365\times10^{-28}\,kg$

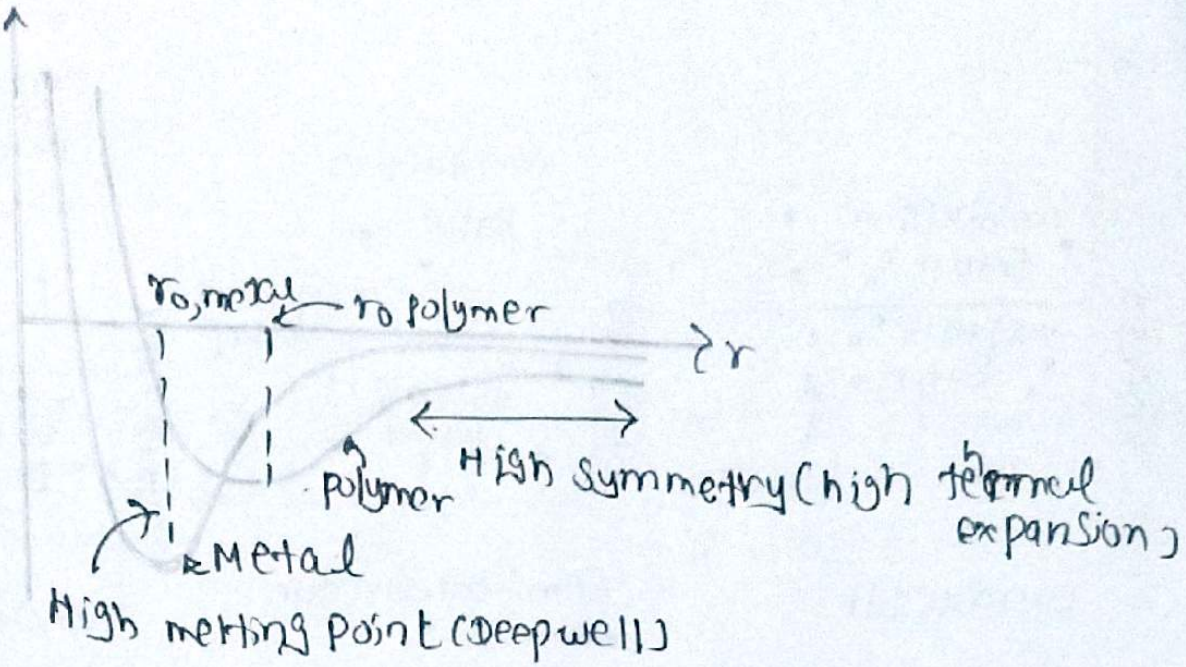
$I = \mu r^2$

$r^2 = \frac{I}{\mu} = \frac{4.60\times10^{-48}}{8.365\times10^{-28}} \approx 5.50\times10^{-21}\,m^2$

$r \approx 7.42\times10^{-11}\,m$

EXAM-2020/2021]

(a) (i)



(ii) (I) once the bond is formed, ions are in their equilibrium distance, $F_{net} = 0$, Repulsive and attractive force perfectly balance out each other.

(II) Attractive force:

$$F_{att} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r_0^2}$$

(iii) (I) H-Cl : $r = 1.30 \times 10^{-10}\ m$

$H \Rightarrow e^+ = 1.602 \times 10^{-19}$, $Cl \Rightarrow e^- = -1.602 \times 10^{-19}\ C$

$$F_{att} = \frac{1}{4\pi\varepsilon_0} \frac{|q_1 \cdot q_2|}{r_0^2}$$

$$= (8.988 \times 10^9) \frac{(1.602 \times 10^{-19})^2}{(1.30 \times 10^{-10})^2}$$

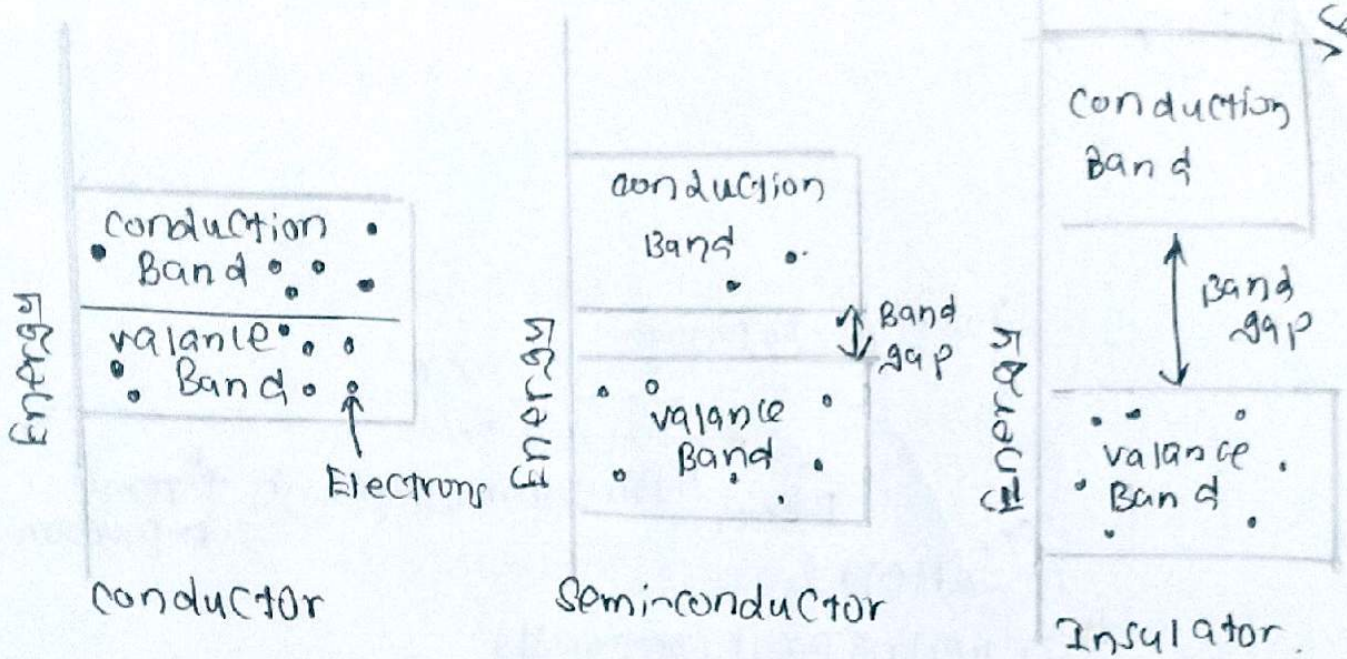
$$F_{att} \approx 1.365 \times 10^{-8}\ N$$

(II) $|F_{rep}| = |F_{att}|$

$$\frac{B}{r_0^7} = 1.365 \times 10^{-8} \Rightarrow B = 1.365 \times 10^{-8} (1.30 \times 10^{-10})^7$$

$$B \approx 8.365 \times 10^{-78}\ Nm^7$$

(B) Band structure Diagrams



- overlap of bands. Electons move freely & high conductivity
- small energy gap Electons can be thermally excited at room temperature
- very large energy gap. Electrons cant be excited across, resulting in extremely low conductivity.

(08) (a) $r_0 = 0.750 \times 10^{-10} m$ $m_H = 1.67 \times 10^{-27} kg$

(i) $E_J = \frac{h^2}{8\pi^2 I} J(J+1)$ $\therefore J = 1$

$E_1 = \frac{h^2}{4\pi^2 I}$

$I = \mu r^2$ & $\mu = \frac{m \cdot m}{2m} = \frac{m}{2} = \frac{1.67 \times 10^{-27} kg}{2} = 0.835 \times 10^{-27} kg$

$I = (0.835 \times 10^{-27})(0.750 \times 10^{-10})^2$

$I \approx 4.697 \times 10^{-48} kgm^2$

$E_1 = \frac{(6.626 \times 10^{-34})^2}{4\pi^2 (4.697 \times 10^{-48})} \approx 2.368 \times 10^{-21} J$

$J = 1 \rightarrow J = 0$

$\Delta E = E_1 - E_0 = \frac{h^2}{4\pi^2 I} - 0$

$= 2.368 \times 10^{-21} J$

$\Delta E = \frac{hc}{\lambda}$

$\lambda = \frac{hc}{2.368 \times 10^{-21}} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{2.368 \times 10^{-21}}$

$\lambda \approx 8.395 \times 10^{-5} m$

(ii) Deuterium (D) has heavier atomic mass than H (But seperation remains the same) so It has ↑ higher μ than H. which results I↑ and $E \downarrow \propto \frac{1}{I} \uparrow$ $E \downarrow \propto \frac{1}{\lambda} \uparrow$, so D has longer wavelength.

(b) (i) $E_n = \frac{h}{2\pi}\sqrt{\frac{K}{\mu}}\left(n + \frac{1}{2}\right)$

$E_{n+1} - E_n = \frac{h}{2\pi}\sqrt{\frac{K}{\mu}}\left(n + 1 + \frac{1}{2} - \left(n + \frac{1}{2}\right)\right)$

$\Delta E = \frac{h}{2\pi}\sqrt{\frac{K}{\mu}}$

E_n = Energy of the nth vibrational state

n = Vibrational quantum number

h = planck's constant

K = Effective Spring constant

μ = Reduced mass of diatomic molecule.

(ii) $K_{HI} = 320 \, Nm^{-1}$ $K_{HF} = 970 \, Nm^{-1}$

$m_I = 10^{-25} kg$ $m_F = 10^{-26} kg$ $m_H = 10^{-27} kg$

$\mu_{HI} = \frac{(10^{-27})(10^{-25})}{10^{-25} + 10^{-27}} \approx \frac{10^{-52}}{10^{-25}} \approx 10^{-27} kg$

$\mu_{HF} = \frac{10^{-26}(10^{-27})}{10^{-25} + 10^{-26}} \approx \frac{10^{-53}}{1.1 \times 10^{-26}} \approx 0.909 \times 10^{-27} kg$

$E \propto \sqrt{\frac{K}{\mu}}$ $\mu_{HI} \approx 10^{-27}$ kg $K_{HI} = 320$ Nm^{-1}

$\mu_{HF} \approx 0.909 \times 10^{-27}$ kg $K_{HF} = 970$ Nm^{-1}

In HF it has lower $\mu \downarrow$ and higher $K \uparrow$; whereas, HI has higher $\mu \uparrow$ and lower $K \downarrow$ so $E_{HI} > E_{HF}$

So HF absorb higher energy photons.

[END EXAM-2019/2020]

(07) (a) (i) Attractive potential : $-\frac{a}{r^m}$

Dominates at longer ranges. Represents van der waals forces. London forces caused by interacting instantaneous dipole moment.

Repulsive potential : $\frac{b}{r^n}$

Dominates at short ranges. Arises from pauli exclusion principle. As atoms forced to close together their electron overlap strong repulsive force.

(ii) $V(r) = -\frac{a}{r^m} + \frac{b}{r^n}$

$$\left.\frac{dV}{dr}\right|_{r=r_0} = 0 \Rightarrow \frac{dV}{dr} = -a(m) r^{-m-1} + b(-n) r^{-n-1}$$

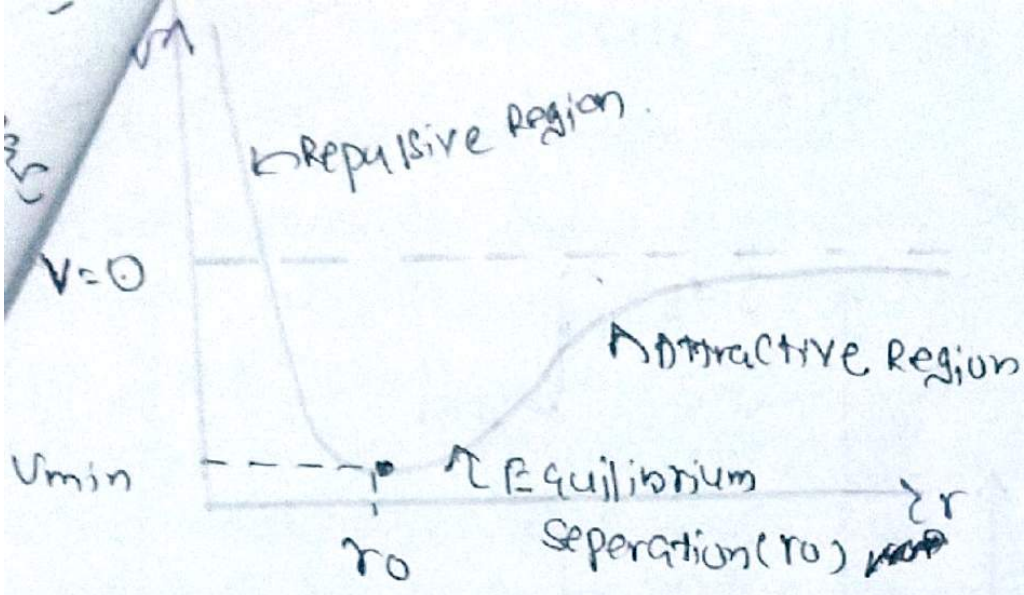
$$0 = am\, r^{-m-1} - bn\, r^{-n-1}$$

$$a m r_0^{-m-1} = bn\, r_0^{-n-1}$$

$$\left(\frac{r^{-m-1}}{r^{-n-1}}\right) = \left(\frac{bn}{am}\right)$$

$$\frac{r_0^{-m} \cdot r_0^{-1}}{r_0^{-n} \cdot r_0^{-1}} = \frac{b^n}{a^m}$$

$$r_0^{(n-m)} = \left(\frac{b^n}{b^m}\right) \Rightarrow r_0 = \left(\frac{b^n}{am}\right)^{\frac{1}{(n-m)}}$$



(b) (i) $L = \sqrt{j(j+1)}\hbar \quad (j = 0, 1, 2 \ldots)$

$E_{rot} = \frac{1}{2} I\omega^2 \quad \because L = I\omega$

$= \frac{1}{2} I \left(\frac{L}{I}\right)^2 = \frac{1}{2} \frac{L^2}{I}$

$= \frac{1}{2} \frac{j(j+1)\hbar^2}{I} \Rightarrow E_{rot} = \frac{j(j+1)\hbar^2}{2I}$

(ii) $\Delta E_{0\to1}$ and $\Delta E_{2\to3}$

$\Delta E_{0\to1} = \frac{\hbar^2}{2I}(1)(1+1) - \frac{\hbar^2}{2I}0(0+1)$

$= \frac{\hbar^2}{I} \Rightarrow \frac{\hbar^2}{I} = hf_1$

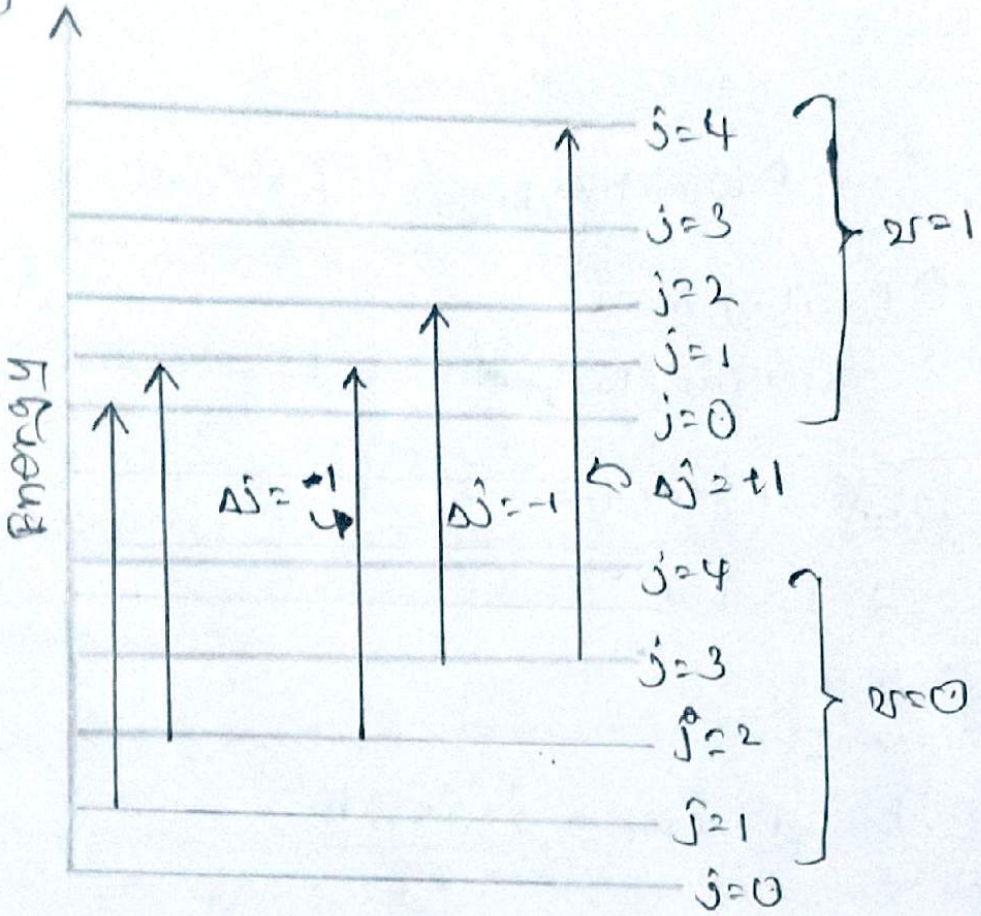
$\Delta E_{2\to3} = \frac{\hbar^2}{2I}(3)(4) - \frac{\hbar^2}{2I}(2)(3)$

$= \frac{3\hbar^2}{I} \Rightarrow \frac{3\hbar^2}{I} = hf_2$

$\frac{f_1}{f_2} = \frac{1}{3}$

(a)(8) $E_{vj} = (v+\frac{1}{2})hf + \frac{h^2}{8\pi^2 I} j(j+1)$

(a)(b)



(c) (i) $E(0,0) = (0+\frac{1}{2})hf + \frac{h^2}{2I}(0)(0+1)$

$= \frac{1}{2}hf$

$E(1,1) = \frac{3}{2}hf + \frac{h^2}{2I}(2)$

$= \frac{3}{2}hf + \frac{h^2}{I}$

(ii) $\Delta E = (E_{1,1}) - (E_{0,0})$

$= \frac{3}{2}hf + \frac{h^2}{I} - \frac{1}{2}hf$

$\Delta E = hf + \frac{h^2}{I}$

[EXAM - 2018/2019]

(a) (i) Water : Hydrogen bonds
Liquid methane: London dispersion forces
Tungsten : Metallic bonds
MgO: Ionic bonds
Diamond: Covalent network bonding
Silicon : Covalent network bonding
Liquid HF: Hydrogen bonding
Brass : Metallic bonding

(ii) Dispersion forces:

(1) Dipole - Dipole (Keesom forces)
- Attractive forces between molecules with permanant dipole moments

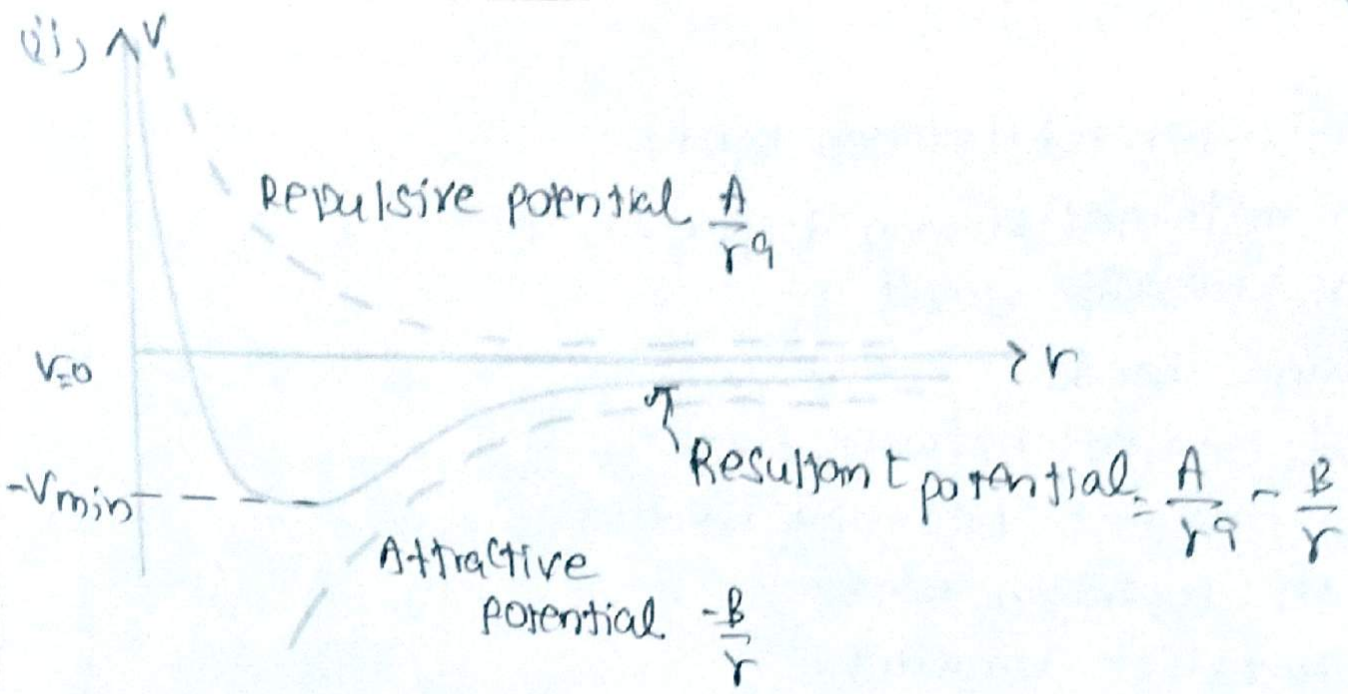
(2) Dipole - induced Dipole (Debye forces)
- When Molecule with a permanant dipole approaches a non-polar molecule.

(3) London
- weak, temporarily present in all atoms & molecules. Arise from instantaneous fluctuations in electron distribution.

(b) (i) Repulsive potential : $\frac{+A}{r^9}$
origin: Dominates at short ranges, Arises from pauli exclusion principle. As atoms forced to close together their electron clouds overlap ⇒ strong repulsive force

Attractive Potential: $\frac{-B}{r}$
origin: Dominates at longer ranges, Represents van der waals forces. London forces are caused by instantaneous dipole moment.

(ii)



(iii) $E_0 = 4.26\ eV \qquad r_0 = 0.236\ nm$

$V(r_0) = -E_0 = -4.26\ eV$

So:

$$\frac{A}{r_0^9} - \frac{B}{r_0} = -4.26\ eV$$

$$\left.\frac{dV}{dr}\right|_{r=r_0} = A(-9)r^{-10} - B(-1)(r^{-2})$$

$$0 = -9Ar^{-10} + Br^{-2}$$

$$\frac{9A}{r_0^{10}} = \frac{B}{r_0^2} \Rightarrow \frac{9A}{B} = r_0^8 \Rightarrow A = \frac{Br_0^8}{9}$$

$$\left(\frac{Br_0^8}{9}\right)\frac{1}{(r_0^9)} - \frac{B}{r_0} = -4.26 \Rightarrow \frac{B}{9r_0} - \frac{B}{r_0} = -4.26$$

$$-\frac{8B}{r_0} = -4.26$$

$$\frac{8B}{0.236\times10^{-9}} = 4.26$$

$$B \approx 1.131\ eV\,nm$$

$$A = \frac{1.131\times(0.238)^8}{9} \approx 1.250\times10^{-6}\ eV\,nm^9$$

$$K = \frac{d^2U}{dr^2}\Big|_{r=r_0}$$

$$= -9Ar^{-10}(-10) + (-2)Br^{-3}$$

$$= 90Ar^{-11} - 2Br^{-3}$$

$$= 90\left(\frac{Br_0^8}{9}\right)r_0^{-11} - 2Br_0^{-3}$$

$$K = 10Br_0^{-3} - 2Br_0^{-3}$$

$$= \frac{8B}{r_0^3} = \frac{8 \times 1.131\ eV\,nm}{(0.236\,nm)^3} \approx 688.4\ eV/nm^2$$

$$1\ \frac{eV}{nm^2} = \frac{1.602 \times 10^{-19}\ J}{(10^{-9}\,m)^2} = 0.1602\ J/m^2 = 0.1602\ N/m$$

$$K = 688.4 \times 0.1602 \approx 110.3\ Nm^{-1}$$

(0.8) (a) (i) $E_j = \frac{\hbar^2}{2I}(j)(j+1)$

j: Rotational Quantum number
I: moment of inertia of the diatomic molecule.

(ii) $\hbar \rightarrow J.s$, $I \rightarrow kgm^2$

$(\hbar)^2 \rightarrow J^2s^2$

$1J = 1\ kg\,m^2/s^2 \Rightarrow (\hbar^2)\ (kg\,m^2s^{-2})^2 s^2$

$kg^2m^4s^{-4}s^2 = kg^2m^4s^{-2}$

$$\text{unit of } \frac{\hbar^2}{I} = \frac{kg^2m^4s^{-2}}{kgm^2} = kgm^2s^{-2}$$

(J)

(iii) $E = 2.48 \times 10^{-4}\ eV$

$= (2.48 \times 10^{-4}\ eV) \times (1.602 \times 10^{-19}\ J/eV)$

$\approx 3.973 \times 10^{-23}\ J$

$\frac{\hbar^2}{I} = E$

$I = \frac{\hbar^2}{E} = \frac{(1.055 \times 10^{-34}\,Js)^2}{(3.773 \times 10^{-23}\,J)} \approx 2.80 \times 10^{-46}\,Kg\,m^2$

$I = \mu r^2 \Rightarrow r = \sqrt{\frac{I}{\mu}}$

Mass of one N = $\frac{14.01 \times 10^{-3}\,Kg/mol}{6.022 \times 10^{23}\,atoms/mol} \approx 2.326 \times 10^{-26}\,Kg$

$\mu = \frac{m \cdot m}{m+m} = \frac{m}{2} \approx 1.163 \times 10^{-26}\,Kg$

$r = \sqrt{\frac{2.80 \times 10^{-46}}{1.163 \times 10^{-26}}} = 0.155\,nm$

(iv) From $J=1$ to $J=0$

$\Delta E = E_1 - E_0 = \frac{\hbar^2}{2I}(1)(2) - 0$

$= \frac{\hbar^2}{I}$

CO $\Rightarrow \mu_{CO} \propto \frac{12 \times 16}{12+16} \approx 6.86$ NO $\Rightarrow \mu_{NO} \propto \frac{14 \times 16}{14+16} \approx 7.47$

$I \propto \mu$ r is the same

$\lambda \propto \frac{1}{E} \Rightarrow E \propto \frac{1}{I} \Rightarrow \lambda \propto I \Rightarrow \lambda \propto \mu$

So NO has higher wavelength (longer) $= 4.5\,nm$

CO has shorter wavelength $= 2.6\,nm$

$\lambda = 2300 \times 10^{-9}\ m$

$\Delta E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^{8})}{(2300 \times 10^{-9})}$

$\Delta E \approx 8.64 \times 10^{-20} J$

(ii) $E_{vib} = (v + 1/2) hf$

Ground state $E_0 = (0 + 1/2) hf = \frac{hf}{2}$

$E_0 = \frac{1}{2} \Delta E \approx 4.32 \times 10^{-20} J$

(iii) $f \propto \sqrt{k}$ $\left(f = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} \right)$

C=C has stronger, stiffer bonds than C-C : so the force constant of C=C is higher than C-C.
So the frequency of C=C is higher than C-C

$f_{C=C} > f_{C-C}$

(iv) Application of atomic vibration

- IR spectroscopy

[END EXAM - 2017/2018]

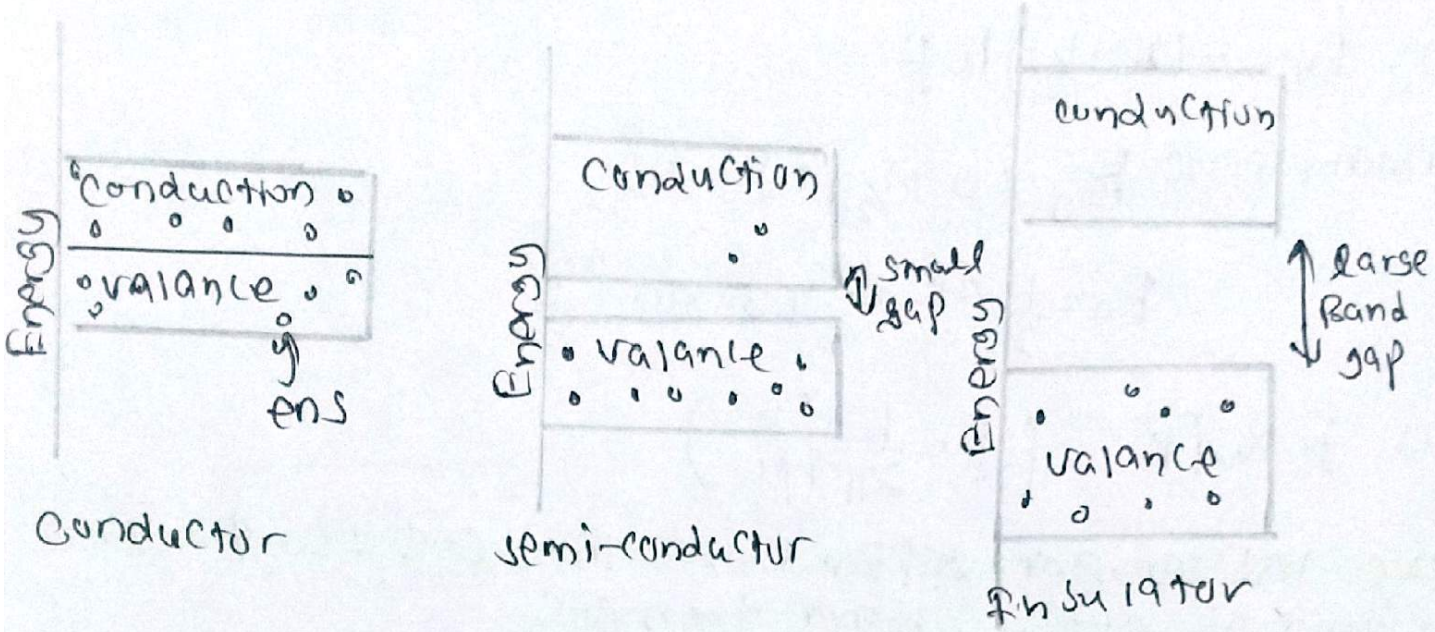
(07)(a)(i) cry stalline state: Atoms, ions or molecules are arranged in a highly ordered, repeating 3-D lattice structure that extends over long distances

Amorphous state: lacks any long-range structural order, arrangement: relatively random and disordered.

(ii) conductors: The valance band is either partially filled or it completely overlaps with the empty conduction band. No energy gap, en free to move and conduct electricity

Semiconductors: Full valance band and empty co
-on (almost) band : small gap. At Absolute zero act as insulayors, at room tempt conducts electricity

Insulators: valance band completely full, conduction band is empty, very large gap, no low conductivity.



(iii) van der waals

(1) Dipole-Dipole, (2) Dipole-induced Dipole, (3) London

(1)
- Dipole-Dipole (Keesom forces)
- molecules with permanant dipolemoments ~~no~~ HCl

(2)
- Dipole-induced Dipole (debye forces)
- molecule with permanant dipole approaches a non-polar ~~no~~ HCl reacting with Ar

(3) London
weak, temporarily present in all atoms & molecules. Arise from instantaneous fluctuations in electron distribution.

(b) (2) Attractive potential $= -\frac{9.00}{r}$

Repulsive potential $= \frac{2.56 \times 10^{-6}}{r^9}$

$\frac{2.56\times10^{-6}}{r^8}$ must +ve eV

So 2.56×10^{-6} must +ve eV nm

$\frac{-9.00}{r}$ eV

so -9.00 must +ve eV.nm

(iii) $\left.\frac{dV}{dr}\right|_{r=r_0} \Rightarrow \frac{dV}{dr} = -9.r^{-10}(2.56\times10^{-6}) - (-1)r^{-2}9.00$

$0 = -9\left(\frac{2.56\times10^{-6}}{r^{10}}\right) + \frac{9.00}{r^2}$

$\frac{9}{r_0^{10}}(2.56\times10^{-6}) = \frac{9}{r_0^2}$

$2.56\times10^{-6} = r_0^8 \Rightarrow r_0 = (2.56\times10^{-6})^{1/8}$

$= 2\times10^{-1}$ nm

(iv) $V(r_0) = \frac{2.56\times10^{-6}}{(0.20)^8} - \frac{9.00}{0.20}$

$= \frac{}{(0.20)^8}$

$= \frac{(0.20)^8}{(0.20)^7} - \frac{9.00}{0.20} = \frac{-8.00}{0.20}$

$= -40.00$ eV

(v) $K = \left.\frac{d^2V}{dr^2}\right|_{r=r_0}$

$K = -9.(-10)r^{-11}(2.56\times10^{-6}) + r^{-3}(-2)9.00$

$= 90\,r^{-11}(2.56\times10^{-6}) - 18.00\,r^{-3}$

$$K = \frac{230.4\times10^{-6}}{(0.20)^{11}} - \frac{18.00}{(0.20)^3}$$

$$= \frac{230.4\times10^{-6}}{(0.20)^8(0.20)^3} - \frac{18.00}{(0.20)^3}$$

$$= \frac{1}{(0.20)^3}\left[\frac{230.4\times10^{-6}}{2.56\times10^{-8}} - 18.00\right] = 9000\ eV/nm^2$$

$$\frac{1\ eV}{nm^2} = \frac{1.602\times10^{-19}\ J}{(10^{-9}\ m)^2} = 0.1602\ J/m^2$$

$$K = 1441.8\ N/m$$

(8) (a) (i) Rotational spectrum provides the frequencies of photons absorbed or emitted during transitions between energy levels. By Identifying $\Delta E \rightarrow I \rightarrow \mu$ μ of the molecule. Also since $I = \mu r^2$ we can know about r.

(ii) $r_0 = 0.13nm$, $j = 1$

$$L = \sqrt{j(j+1)}\,\hbar = \sqrt{2}\,\hbar \qquad L = I\omega$$

$$\sqrt{2}\hbar = I\omega \Rightarrow \omega = \frac{\sqrt{2}\hbar}{I}$$

$$I = \mu r^2$$

$$\mu = \frac{m_H \cdot m_{Cl}}{m_H + m_{Cl}} = \frac{(1.04)(35.04)}{(1.04)+(35.04)} \times 0.9722u$$

$$\mu \approx 0.9722 \times 1.66\times10^{-27}\ kg$$

$$\approx 1.614\times10^{-27}\ kg$$

$I = \mu r^2$

$= 1.614 \times 10^{-27} \times (0.13 \times 10^{-9})^2$

$\approx 2.728 \times 10^{-47}\ kgm^2$

$\omega = \sqrt{\frac{\Sigma}{2\pi (2.728 \times 10^{-47})}}\ \hbar \approx 5.47 \times 10^{12}\ rad/s$

(b) (i) $\Sigma(HCl) = 428\ KJmol^{-1}$ $NaCl = 410\ KJmol^{-1}$

$f = \frac{1}{2\pi}\sqrt{\frac{K}{\mu}}$

Higher bond strength → Stiff K ↑

$f \propto \sqrt{K}$ so HCl has high frequency

when comparing μ : $\mu_{HCl} \approx 1u$ $\mu_{NaCl} \approx 14u$

so HCl K ↑ NaCl K ↓ since $f \propto \sqrt{\frac{K}{\mu}}$

μ ↓ μ ↑

HCl has higher frequency.

(ii) $\Delta E = hf$

HCl has high f than that of NaCl

$\Delta E_{HCl} > \Delta E_{NaCl}$

$\lambda_{HCl} < \lambda_{NaCl}$

↗ ↑

near infrared far infrared

(mid to near)

(iii) $\lambda_{HCl} = 3760 \times 10^{-9}\ m$ ($\nu = 0 \rightarrow \nu = 1$)

$\Delta E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{3760 \times 10^{-9}}$

$\approx 5.287 \times 10^{-20}\ J$

$$f = \frac{\Delta E}{h} = \frac{5.287 \times 10^{-20}}{6.626 \times 10^{-34}} \approx 7.979 \times 10^{13}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{K}{\mu}} \Rightarrow 4\pi^2 f \mu = K$$

$$4\pi^2 (7.979 \times 10^{13}) \times 1.614 \times 10^{-27} = K$$

$$K \approx 405.6 \text{ N/m}$$